SPLITTING OFF THE VANISHING LOCUS OF ONE SECTION WHILE KEEPING THE CODIMENSION BOUND

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The situation fixing the sheaf of modules, the finite dimensional spaces of values and the sections, whose fields and dimension function the statement uses. used from the statement at line 38811 onward. Fixes the notation M, V(x), s_1 to s_h and F so that the statement and the proof talk about the same objects.
- For the case of the base value of the induction one can prescribe the values of a section at finitely many points. invoked at proof lines 38835-38837. Produces the section s and the choice F' equal to the whole space in the case k = 0.
- A finite set of functions can be prescribed at one finite set of points and made to vanish at another, by the Chinese remainder argument for the ring of functions. invoked at proof lines 38866-38868. Produces the element f equal to one at the y_j and zero at the z_l, which builds the perturbed section u plus f t.
- The codimension function on a Noetherian topological space and the finiteness of the number of irreducible components. used throughout the proof from lines 38831-38835 and at many later steps. Lets the proof decompose the closed sets G and H into irreducible components, pick points outside the other components, and compare codimensions.

Source reference. Original lemma 0GVA: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38795–38795. Used at proof lines 38835-38837 for the base case of the induction, where the whole space is taken as the closed set so no codimension estimate is needed; every other step of the induction is inside the counted span.

Source reference. Original lemma 00DT: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 38866-38868 to produce the element f taking value one at the chosen points of the first closed set and zero at those of the second; the construction of the perturbed section and its verification are inside the counted span.

Source reference. Original section 0050: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 38832-38835 as the source of the finiteness and decomposition statements for closed sets; the decompositions into irreducible components used later are performed inside the counted span.

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Source reference. Original section 02I0: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 38831-38833 to point at the definition of codimension used in the statement and in the comparisons; all codimension comparisons are made inside the counted span.

Proposition 0.2. *In Situation 0.3 assume Ω is a Noetherian topological space. Let $s_1, \ldots, s_h \in M$. Let $Z(s_1, \ldots, s_h) \subset F \subset \Omega$ be closed. Let $x_1, \ldots, x_n \in F$ be pairwise distinct. Let $v_i \in V(x_i)$. Let $k \geq 0$ be an integer such that*

$$(*) \quad h + k \leq \dim_{\kappa(x)} V(x) \text{ for all } x \in \Omega$$

Then there exist $s \in M$ and $F' \subset \Omega$ closed such that

(a) *$s(x_i)$ maps to v_i,*
(b) *$Z(s_1, \ldots, s_h, s) \subset F \cup F'$, and*
(c) *every irreducible component of F' has codimension $\geq k$ in Ω.*

Situation 0.3. Here R is a ring and M is a finitely presented R-module. Denote $\Omega \subset \mathrm{Spec}(R)$ the set of closed points with the induced topology. For $x \in \Omega$ denote $M(x) = M/xM$ the fibre of M at x. This is a finite dimensional vector space over the residue field $\kappa(x)$ at x. Given $s \in M$ we denote $s(x)$ the image of s in $M(x)$.

Proof. We note that codimension was defined in Topology, Section 02I0 and that we will use some results on Noetherian topological spaces contained in Topology, Section 0050.

The proof is by induction on k. If $k = 0$, then we choose $s \in M$ as in Lemma 0GVA and we choose $F' = \Omega$.

Assume $k > 0$. By our induction hypothesis we may choose $u \in M$ and $G \subset \Omega$ closed satisfying (a), (b), (c) for $s_1, \ldots, s_h$, F, $x_1, \ldots, x_n$, $v_1, \ldots, v_n$, and $k - 1$.

Let $G = G_1 \cup \ldots \cup G_m$ be the decomposition of G into its irreducible components. If $G_j \subset F$, then we can remove it from the list. Thus we may assume G_j is not contained in F for $j = 1, \ldots, m$. For $j = 1, \ldots, m$ choose $y_j \in G_j$ with $y_j \notin F$ and $y_j \notin G_{j'}$ for $j' \neq j$. This is possible as there are no inclusions among the irreducible components of G. Choose $w_j \in V(y_j)$ not contained in the span of the images of $s_1(y_j), \ldots, s_h(y_j)$; this is possible because $h + k \leq \dim V(y_j)$ and $k > 0$.

Apply the induction hypothesis to the $h + 1$ sections $s_1, \ldots, s_h, u$, the closed set $F \cup G$, the points $x_1, \ldots, x_n, y_1, \ldots, y_m \in F \cup G$, the elements $0 \in V(x_i)$ and $w_j \in V(y_j)$, and the integer $k - 1$. Note that we have increased h by 1 and decreased k by 1 hence the assumption $(*)$ of the proposition remains valid. This produces $t \in M$ and $H \subset \Omega$ closed satisfying (a), (b), (c) for $s_1, \ldots, s_h, u$, $F \cup G$, $x_1, \ldots, x_n, y_1, \ldots, y_m$, $0, \ldots, 0, w_1, \ldots, w_m$, and $k - 1$.

Let $H_1, \ldots, H_p \subset H$ be the irreducible components of H which are not contained in $F \cup G$. As before pick $z_l \in H_l$, $z_l \notin F \cup G$ and $z_l \notin H_{l'}$ for $l' \neq l$. Using Algebra, Lemma 00DT we may choose $f \in R$ such that $f(y_j) = 1$, $j = 1, \ldots, m$ and $f(z_l) = 0$, $l = 1, \ldots, p$. Claim: the element $s = u + ft$ works.

First, the value $s(x_i)$ agrees with $u(x_i)$ because $t(x_i) = 0$ and hence we see that $s(x_i)$ maps to v_i. This proves (a). To finish the proof it suffices to show that every irreducible component Z of $Z(s_1, \ldots, s_h, s)$ not contained in F has codimension $\geq k$ in Ω. Namely, then we can set F' equal to the union of these and we get (b) and (c). We can see that irreducible components Z of $Z(s_1, \ldots, s_h, s)$ of codimension $\leq k - 1$ do not exist as follows:

(1) Observe that $Z(s_1, \ldots, s_h, s) \subset Z(s_1, \ldots, s_h, u, t) = F \cup H$ as $s = u + ft$. Hence $Z \subset H$.

(2) The irreducible components of H have codimension $\geq k-1$. Hence Z is equal to an irreducible component of H as Z has codimension $\leq k-1$. Hence $Z = H_l$ for some $l \in \{1, \ldots, p\}$ or $Z = G_j$ for some $j \in \{1, \ldots m\}$.
(3) But $Z = G_j$ is impossible as $s_1(y_j), \ldots, s_h(y_j)$ map to linearly independent elements of $V(y_j)$ and $s(y_j) = u(y_j) + f(y_j)t(y_j) = u(y_j) + t(y_j)$ maps to an element of the form

$$\text{linear combination images of } s_i(y_j) + w_j$$

which is linearly independent of the images of $s_1(y_j), \ldots, s_h(y_j)$ in $V(y_j)$ by our choice of w_j.
(4) Also $Z = H_l$ is impossible. Namely, again $s_1(z_l), \ldots, s_h(z_l)$ map to linearly independent elements of $V(z_l)$ and $s(z_l) = u(z_l) + f(z_l)t(z_l) = u(z_l)$ maps to an element of $V(z_l)$ linearly independent of those as $z_l \notin F \cup G$.

This finishes the proof. □